Mid Sweden University (Mittuniversitetet)

Omnisound

User-Centred Design and Evaluation
of a Unified Audio Interface

DT168G / IU127G
Human-Computer Interaction / Interaction Design

Project group: Group 1

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Abstract

TODO: Write an abstract of approximately 250 words after the results and discussion are complete. It should summarise the project's background, purpose, method, results and conclusions.

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Chapter 1

Introduction

1.1 Background

Music, podcasts and audiobooks can form part of the same person's listening activities, while being accessed through different applications. In such a situation, finding audio involves choosing both what to listen to and where to look for it. A listener may save an item in one application, continue a different item elsewhere and later need to remember which application holds each one. These possible organisational difficulties motivate the project.

Finding content is only one part of this problem. Saving an interesting item should also make it possible to find that item again. Similarly, returning to an unfinished podcast or audiobook requires a recognisable route back to the content and its listening position. For example, a student who stops listening to an audiobook during a study break should be able to identify the unfinished book and continue it later.

Combining the three audio types could provide a common place for these activities, but it also raises a design question. A mixed collection may be difficult to navigate if its content types are unclear or if too many items compete for attention. Music tracks, podcast episodes and audiobook chapters also need to remain identifiable within a shared interface. Common navigation and saving controls therefore have to be balanced with clear categories and information about each item.

Existing services already take different approaches to this organisation. Spotify combines music, podcasts and audiobooks in one service, while keeping the content types identifiable within its listening experience [1]. Apple offers separate applications for music, podcasts and audiobooks through Apple Music, Apple Podcasts and Apple Books [2]. These examples show that combining audio content is not a new idea, and neither approach is automatically easier for every user. Omnisound instead explores a specific interface question: whether shared search, a common library and clear content categories can make it easier to find saved material and continue unfinished listening without confusing the different audio types.

Omnisound is the project's response to this question: a web-interface prototype for finding, saving and resuming audio in one organised place. It primarily targets students

who regularly listen to music, podcasts or audiobooks, including listeners who use several content types and those who mainly listen to music. The project investigates how the interface can help these listeners keep track of saved and unfinished content.

The initial high-fidelity prototype (V1) is designed primarily for a desktop browser. It provides a shared search, content-type filters, a library of saved items and a Continue Listening area. Its two main task flows are finding and saving an item, then locating it in the library, and resuming an unfinished podcast or audiobook. These flows give the project a specific interaction focus within the broader audio concept.

1.2 Purpose

The purpose of the project is to investigate how a unified audio interface can support finding, organising and resuming different types of audio content while keeping the experience clear and manageable for the intended users. The motivation is to explore whether a common search and library can reduce the effort of returning to wanted content without making the differences between music, podcasts and audiobooks harder to understand.

This investigation also provides an opportunity to apply user-centred design within the HCI course. The course project requires an iterative process and usability testing in at least one iteration [3, pp. 2–3]. For Omnisound, the purpose is therefore pursued through requirements, alternative designs, prototypes and evaluation. The desired qualities are ease of learning, clear organisation and useful feedback; whether the design supports these qualities must be assessed through the project work.

1.3 Goals

The completed project should provide an identified set of user needs and a prioritised set of interface requirements. The requirements should make clear which listening activities the interface is intended to support and how they relate to the information collected from the target group. This provides a basis for assessing the design beyond its visual appearance.

The design deliverables should include documented alternatives, low-fidelity sketches and interactive high-fidelity prototypes. Alternatives should cover the organisation of audio categories and the presentation of content, such as category tabs compared with a mixed view with filters, and cards compared with lists. The sketches should preserve the early ideas, while the V1 and V2 prototypes should make the selected task flows available for examination. Together, these materials should show how the interface developed and why particular solutions were chosen.

A further deliverable is a documented usability evaluation of the important tasks: finding and saving content, locating saved content and returning to unfinished spoken audio. The evaluation should provide observations of task completion and difficulties, together with an account of how those observations informed design decisions. A record

of design iterations and reasoned conclusions should connect the requirements, prototypes and evaluation. These goals concern identifiable project outputs; they do not assume that all proposed changes will prove effective.

1.4 Delimitations

Omnisound is an interface prototype. The project focuses on navigation, content organisation and the visibility of actions and listening progress. It does not implement music-streaming infrastructure, integration with commercial audio services, payment or subscription processing, licensing systems, backend databases or production account management. The prototype is implemented only as a front end. Limiting the implementation in this way allows the design work to concentrate on the selected interactions within the course project.

Search uses a sample catalogue, and the player controls and displayed progress represent the intended listening experience. They are sufficient to explore how users move through the interface, recognise content and interpret feedback. They do not demonstrate reliable audio delivery, synchronisation between devices or the operation of a real streaming service. Conclusions must therefore concern interaction with the prototype rather than the technical performance of a complete product.

The task scope is also limited. Search and save, retrieval from the library and continuation of unfinished podcasts or audiobooks receive priority. Other possible activities, such as managing a production account or exploring a large commercial catalogue, are outside this evaluation focus. The primary context is desktop use, so the project cannot establish usability across all mobile devices or listening environments. Likewise, conclusions drawn from evaluation with the intended student group should be considered in relation to that group and the tasks tested.

Following the first usability test, a second version of Omnisound was developed to address the problems identified during evaluation. The changes focused on reducing the size of the Home banner, making Continue Listening easier to discover, and improving the clarity of the Save/Saved control. A second usability test was then conducted using V2 and the same three core tasks, allowing the group to compare the two iterations. The testing procedures and findings are presented in the Method and Results chapters. Since Omnisound remains an interface prototype, the conclusions concern the tested interactions rather than the technical performance of a complete streaming service.

1.5 Division of Work

The project was carried out by Obid Alrhman Aljabi, Yacoub Dawli and Sarshar Chavoshi Rad.

Yacoub Dawli was responsible for leading the interviews and moderating the usability tests. Obid Alrhman Aljabi and Sarshar Chavoshi Rad were responsible for documenting

participant responses and observations. During usability testing, Obid focused on participant behaviour and comments, while Sarshar documented task completion times and navigation steps.

The group collaborated on analysing user needs, discussing requirements and evaluating different interface designs. The design process included low-fidelity sketches, two high-fidelity prototypes and usability evaluations. Findings from the first evaluation informed the changes introduced in the second prototype.

The project findings and design decisions were discussed within the group and formed the basis for the report.

Chapter 2

Theory

2.1 User-Centred Design and Iteration

User-centred design begins with understanding the people who will use an interface, what they want to achieve and the circumstances in which they will use it. Rogers, Sharp and Preece explain that understanding people's activities and differences helps designers question assumptions about their needs [4, pp. 16–17]. A target group therefore provides a starting point for research, rather than a reason to assume that all its members have the same habits or expectations.

User needs and goals guide the requirements for an interface. Designers can then explore alternative ways to support those goals and represent the ideas in prototypes. Evaluation with intended users examines whether a proposed solution works as expected. A prototype is a candidate solution that can be tested by observing people using it [5, Sec. "What Is a Testable Prototype?"]. This makes evaluation part of design, rather than a check reserved for the finished product.

Iteration connects evaluation to further design decisions. Findings can lead to revised requirements, a different alternative or changes to an existing interface, followed by another evaluation [5, Sec. "Why Test a Prototype?"]. Simply producing several versions does not establish this connection. A meaningful iteration should explain which problem motivated a change and examine whether the problem remains. This principle is relevant to Omnisound's student audience and its two evaluated high-fidelity prototype iterations.

2.2 Usability and User Experience

ISO 9241-11:2018 relates usability to specified users, goals and contexts of use through effectiveness, efficiency and satisfaction. These definitions are reproduced in ISO/TR 25060:2023 [6, Sec. 3.1]. Usability concerns what people can achieve with a system in a particular situation; it cannot be established from appearance alone.

Effectiveness concerns accurate and complete achievement of a goal, while efficiency concerns the resources used in relation to that outcome [6, Secs. 3.1.7–3.1.8]. In an audio

interface, effectiveness could involve successfully saving an item and finding it again. Efficiency concerns the time and effort required to reach that outcome. Successful completion and a straightforward route are thus different aspects of the same task. Satisfaction concerns how well the experience of use meets the user's needs and expectations [6, Sec. 3.1.9].

Learnability and utility are additional useful HCI concepts. Learnability concerns how easily a person can learn to use an interface, including getting started with unfamiliar controls. Utility concerns whether the available functions support what users need to do [4, p. 21]. For example, a clearly labelled search may be easy to learn, while a way to return to unfinished audio addresses a functional need. These concepts complement the three ISO dimensions; they are not additional components of the ISO definition of usability.

User experience includes people's perceptions and responses before, during and after interaction, including anticipated use [6, Sec. 3.1.15]. Satisfaction and frustration are relevant examples. Task completion alone cannot explain how an interaction feels. Together, these concepts provide a basis for examining searching, saving, retrieving and resuming audio without treating one measure as a complete assessment of the interface.

2.3 Interface Design Principles

Visibility means making the options needed for a task apparent to the user [4, p. 28]. An available function has limited value if people cannot discover it. In an audio interface, recognisable navigation and information about listening progress can help users identify where to go and what can be continued.

Feedback communicates the outcome of an action and the current system state [7, Sec. 1]. After saving an item, a visible confirmation can help the user understand that the action succeeded. Feedback should be timely and clear enough to support the next decision, such as continuing to browse or opening the saved collection.

Consistency means using similar terms, controls and behaviour for similar interactions, both within a product and in relation to familiar conventions [7, Sec. 4]. A shared saving action across different audio types can support predictable interaction. Consistency still allows relevant differences in content, such as episode titles and audiobook chapters.

Affordances concern the actions possible in the relationship between a person and an object. Signifiers are perceivable cues that communicate what can be done [8]. In a graphical interface, a button label, icon or visual treatment can signify an available action. A control's appearance and the action it actually supports should agree. This distinction helps explain why making an action available and making it understandable are separate design concerns.

2.4 Cognition and Mental Models

A mental model is a user's understanding of how a system works. It supports predictions about what an action will do and develops through previous experience [9, Sec. "What Are Mental Models?"]. Users and designers may have different models of the same interface. Familiarity with another audio service may, for example, shape expectations about where saved items belong or whether opening unfinished content will resume it. Clear category names and navigation can help communicate how a shared audio collection is organised.

Attention concerns which information people notice while pursuing a goal. Layout and visual prominence matter because competing information can reduce the visibility of relevant content [7, Sec. 8]. A large introductory area and a route back to unfinished listening may compete for attention on the same screen. This provides a reason to examine the relative prominence of elements, without assuming that size alone determines what every user notices.

Recognition over recall means displaying options and information that users can identify, rather than requiring them to remember details [7, Sec. 6]. Visible content categories, item titles and progress information can support this recognition. For an audio collection, these cues may help someone distinguish saved material from unfinished listening and choose a suitable next action.

2.5 User Research and Evaluation Methods

Qualitative interviews explore people's experiences, needs and expectations. Open questions and follow-up questions allow participants to explain what matters to them in their own terms [10]. Interviews can inform requirements, but they collect accounts of behaviour rather than direct observations of interface use. Combining these accounts with observations can provide a fuller basis for understanding users.

Affinity diagramming organises related observations or ideas into groups that express common themes [11, Sec. "Defining Affinity Diagramming"]. It can help a design team make sense of interview notes and identify connections between needs. The grouping is an interpretation of the material, so observations that differ from a common theme should also remain visible in the analysis.

Personas are research-informed representations of users' goals, needs and expectations. Qualitative personas draw on patterns in research to help a team focus its design decisions [12]. They summarise relevant differences between users rather than reproducing every interview as a separate profile. A persona based only on assumptions should not be treated as evidence about the intended audience, and a qualitative persona does not establish how common a user type is in the wider population.

Low-fidelity prototypes, such as paper sketches, represent early ideas with limited detail. They allow designers to explore layouts and navigation before investing in a detailed interface, and can be changed or discarded readily [5, Sec. "Benefits of Low-Fidelity Prototypes"]. More detailed, interactive prototypes allow evaluation of specific

controls and task flows [5, Sec. “Benefits of High-Fidelity Prototypes”]. The appropriate level of detail depends on the question being investigated; visual polish alone does not establish that a design meets users’ needs.

Think-aloud usability testing asks participants to express their thoughts while completing representative tasks with an interface [13, Sec. “Defining Thinking Aloud Testing”]. Observations of actions, hesitation and comments can reveal difficulties and help explain why a participant chose a particular route. These observations can inform redesign, including changes to the visibility or interpretation of controls.

Qualitative usability tests also have limitations. Speaking aloud creates an unusual situation, and prompts can influence behaviour [13, Sec. “Downsides of Think-Aloud”]. Findings from a small group can identify problems in the tested interactions, but do not automatically provide statistically generalisable conclusions. Task times and navigation steps should therefore be interpreted alongside observations and the test context, rather than treated as universal measures of performance.

2.6 Research Ethics

User research should respect participants’ ability to make an informed and voluntary decision. Vetenskapsrådet’s *God forskningssed 2024* explains that people should understand what participation involves, how information will be handled and how to withdraw consent [14, pp. 62–63]. Information should be understandable to the intended participants, who should have an opportunity to ask questions and decline or stop participation without pressure.

Privacy requires responsible handling of participant information. Only information needed for the study should be collected, and it should be protected appropriately. Removing names alone may not prevent identification when other details can be combined [14, pp. 66–67]. This matters when selecting background details, quotations and observations for a report.

Accurate reporting is also an ethical responsibility. Observations and measurements should be presented honestly, with interpretations and limitations made clear [14, pp. 11–12]. Results that challenge a design should be retained rather than adjusted to support a preferred conclusion. These principles provide the basis for describing the group’s documented ethical procedures in the Method chapter.

Chapter 3

Method

3.1 Overall Design Process

The project combined early concept development, design alternatives, prototyping and two rounds of usability testing. The concept and paper sketches established the intended listening activities and possible ways to organise the interface. These ideas were developed into the first high-fidelity prototype, V1. Its evaluation provided a basis for a targeted redesign, which produced V2. A second usability test then examined the updated interface using the same core task goals. Figure 3.1 summarises this sequence.

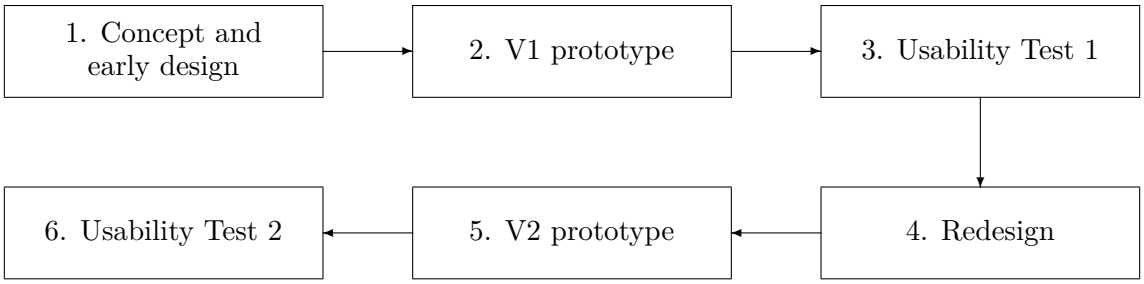


Figure 3.1: Omnisound's design and evaluation process (original project diagram).

This approach was suitable because combining audio types raises questions about how people understand categories, saved content and listening progress. An interactive prototype allowed the group to observe those decisions during use. The user-centred approach described in Section 2.1 therefore informed evaluation of concrete tasks. Iteration connected the observed problems to changes and another evaluation; creating a second version alone would not have provided that connection.

3.2 Initial Design Studio

The early design work was documented through concept material and low-fidelity sketches. The Session 4 concept sheet set out a shared audio interface for students, with finding,

saving and returning to content as its main activities. It also recorded alternative approaches to organisation: separate category tabs compared with a mixed feed and filters, and list views compared with cards. These alternatives framed the question of how to combine content while keeping its different types recognisable.

The original paper sketches explored a navigation area, grouped audio categories and a player layout. Together, they made it possible to examine where navigation, content and playback controls could fit before committing to a detailed interface. They provided concrete material for discussing interface ideas, without requiring each alternative to be fully implemented.

At this stage, the focus was the organisation of activities and information. The concept material connected possible layouts to shared search, a common library and continued listening. Keeping the drawings simple supported consideration of these relationships before visual detail and interactive behaviour became the main focus. The sketches and selected design are presented in the Design chapter.

3.3 Data Collection

The Project Session 2 alternative assignment documented a qualitative interview plan. Its questions concerned existing audio applications, features people liked, possible improvements and expectations of a combined audio service. The DOCX version focused on students who regularly used audio applications and included switching between music, podcasts and audiobooks. The submitted PDF also proposed showing interface ideas and asking about first impressions, possible workflows and missing functions. Both versions specified handwritten notes, with Yacoub Dawli leading the interviews and Obid Alrhman Aljabi and Sarshar Chavoshi Rad recording responses.

Qualitative questions suited the aim of understanding listening habits and expectations in people's own terms [10]. Asking about existing applications could help identify familiar patterns, while questions about a combined service could explore what organisation users expected. These questions provided a planned starting point for user research.

Completed initial interview responses are not available in the retained project material. The plan therefore does not establish an interview sample or findings, and no specific interface requirement is attributed to an unrecorded interview response. The usability-test records do contain participants' background answers and post-test feedback. These documented accounts were collected around interaction with V1 and V2 and provided context for the observed task behaviour.

3.4 Data Analysis and Requirements

The concept sheet and prototype task flows provide the documented basis for the interface requirements. The intended activities were finding audio, saving it in a common collection, retrieving saved material and continuing unfinished listening. Keeping music, podcasts

and audiobooks identifiable was a related organisational requirement. Navigation labels, content-type information, listening progress and clear action feedback supported these activities.

The group concentrated the implementation on paths that could be followed from beginning to end. Finding an item was connected to saving it and checking the Library, while returning to unfinished content was connected to recognising its progress and resuming it. Category browsing provided another way to examine whether the different audio types remained clear. This made the priorities concrete enough to inform both the prototype and its evaluation.

Prioritisation mattered because a course prototype could not implement every feature of a commercial audio service. The selected interactions received attention ahead of account management, subscriptions and streaming infrastructure. The concept's desired qualities, including ease of learning and clear organisation, became questions for usability testing. The later analysis then used recorded task behaviour and feedback to identify where the interface needed refinement. No completed persona or affinity-diagram outputs are documented, so these methods are not treated as completed steps in the requirements analysis.

3.5 Prototyping Methods

Low-fidelity sketches and high-fidelity prototypes served different purposes. The paper drawings explored layouts and relationships between navigation, audio categories and playback. They allowed interface ideas to be considered before the group invested in detailed screens. This use of simple drawings is consistent with the value of low-fidelity prototypes for making design changes with limited preparation [5, Sec. "Benefits of Low-Fidelity Prototypes"].

V1 was a desktop-first, browser-based prototype implemented as a front end. Its main screens were Home, Search, Library, content detail and the full player. These screens supported connected task paths rather than only showing isolated layouts. Users could search the sample catalogue, open an item, change its saved state, find it in the Library and open unfinished spoken audio at its displayed progress. This level of interaction allowed the tests to examine labels, navigation choices, controls and feedback.

The catalogue and listening states represented the intended experience. The prototype did not provide a complete streaming service, so opening the player at an existing position was an interface response, not a test of audio delivery. V2 retained this implementation scope and the main task paths. It provided a revised interface for a second evaluation, while V1 remained available as the earlier version. Detailed screen descriptions and visual comparisons belong in the Design chapter.

3.6 Usability Testing

Two usability-testing rounds were conducted with three participants in each round. P1, P2 and P3 tested V1, while P4, P5 and P6 tested V2. All six were students, aged 20–26, who reported regular use of audio applications. Their backgrounds included mainly music listening and combinations of music, podcasts and audiobooks. These characteristics were relevant to the intended student audience. The records describe participants' listening habits but do not specify how they were recruited.

The evaluations used the interactive prototypes in a browser on a laptop, within the project's desktop-use focus. The V1 plan identifies Mittuniversitetet in Sundsvall as the location and specifies that the prototype ran locally. The V2 plan also specifies a laptop and browser. Participants' usual mobile or computer use was recorded as background information; the sessions did not constitute a comparison between devices.

The group divided moderation and observation between its members. Yacoub Dawli led the sessions and presented the tasks. Obid Alrhman Aljabi documented participant behaviour and comments, while Sarshar Chavoshi Rad recorded task times, clicks and navigation observations. This division allowed the moderator to attend to the participant while the observers recorded the interaction.

Each session began with an introduction to the university project and the purpose of the test. After information about participation and consent, the moderator asked which audio applications the participant normally used, which audio types they listened to, how often they listened and whether they normally used mobile devices, computers or both. The three tasks were then presented individually. Participants were asked to explain what they were looking for, what they expected to happen and anything they found unclear while using the interface. This followed the think-aloud approach introduced in Section 2.5 [13, Sec. "Defining Thinking Aloud Testing"].

The instructions described the intended goal without specifying the control to select. For example, the unfinished-audiobook task did not tell participants to use Continue Listening. This left their choice of route available for observation. When prompting was needed, the moderator asked neutral questions about what the participant was thinking or expected. Participants were encouraged to try what made sense to them instead of being immediately directed to a solution.

Table 3.1 summarises the three core tasks and the completion criteria recorded in the test documentation. The first task included checking that the saved item could be found again. The second concerned returning to an existing listening position, while the third examined access to one content category. These tasks covered the main interaction priorities without requiring every prototype function to be tested.

The same core goals and completion criteria were used in the second round so that the group could compare the two versions without replacing the activities being examined. The wording allowed different valid routes to the goal. Finding content through a category, for example, could still meet the first task's criterion. The records could therefore distinguish task completion from the particular route chosen.

Table 3.1: Core task goals and completion criteria used in both testing rounds.

Task	Goal	Completion criterion
1	Find and save Weezer for later.	The item is saved and can be found in the Library.
2	Find an unfinished audiobook and continue listening.	The player opens at the existing progress position.
3	Find where to browse podcasts.	The Podcasts category is reached.

The task records covered completion, elapsed time in seconds, click counts and navigation paths, alongside notes on hesitation, incorrect actions and any need for help. Relevant think-aloud comments accompanied these observations. Completion related to effectiveness, while time, clicks and routes provided information about effort and efficiency. Hesitation and comments helped examine whether controls were discoverable and whether their feedback was understood. These records supported interpretation of the measured values; a time or click count alone did not explain a difficulty.

Post-test questions covered the overall impression, easy or confusing parts, unclear labels, missing functions and suggested changes. The V2 records also included questions about Save and discoverability. The group reviewed these responses alongside the task observations in each round. This connected participants' impressions to what happened during interaction. Individual accounts, measurements and the comparison of results are presented in the Results chapter.

3.7 Iterative Evaluation

After the first evaluation, the group brought together task records, navigation paths and post-test comments to identify issues for redesign. The V1 analysis linked each identified issue to supporting observations, its effect on the tested task and a proposed action. The prioritised problem list distinguished concerns about reaching or discovering task-related content from smaller concerns about visual clarity. It also identified parts of the interface to retain.

The resulting redesign focused on Continue Listening visibility, the height of the Home introductory area and the clarity of the Save/Saved control. These areas connected the recorded concerns to specific changes that could be examined in another test. The basic navigation, categories, Library and core task goals were kept consistent. This limited the scope of the redesign and retained the main interaction context for comparison.

V2 was then evaluated with new participants using the procedure and tasks described in Section 3.6. Using different participants avoided repeating the tasks with people who had already learned the V1 paths. The second-round analysis considered the same completion criteria, measures, routes and participant feedback, including observations related to the targeted changes.

This comparison concerned the tested prototype interactions. With three different participants per round, differences between the records cannot be attributed solely to the redesign or generalised to all students. The purpose was to examine the revised design against the earlier concerns, without assuming that changing the interface established an improvement.

3.8 Research Ethics

The participant introductions explained that the group was evaluating the interface, not the person using it. They described the university project, the observation and written note-taking, and the intended use of the information. Participants were told that taking part was voluntary, that they could stop at any time and that they could raise questions or discomfort. The moderator then asked whether they were comfortable participating, and the records include agreement before the tasks. This approach addressed informed and voluntary participation as discussed in Section 2.6 and in Vetenskapsrådet's guidance [14, pp. 62–63].

The introductions stated that written notes would be anonymous and used for the university project. The retained records identify participants as P1–P6 instead of using personal names. Background information about listening habits and device use provided context for the tests without requiring names in the report.

Accurate reporting also informed the treatment of the material. The testing records distinguish observations and participant comments from the group's interpretations. This distinction is maintained when describing the method and reporting the results, consistent with the responsibility to communicate research honestly [14, pp. 11–12].

Chapter 4

Design

TODO: Chapter not yet written.

Chapter 5

Results

TODO: Chapter not yet written.

Chapter 6

Discussion

TODO: Chapter not yet written.

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